

# Compare Ratios

Lesson 3-5

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*\$3 per 1 pound → unit rate is \$3/lb***Unit rate**

A rate for just 1 of something, like cost for 1 item.

*4:6 and 2:3 are equivalent (both simplify to 2:3)***Equivalent ratio**

Two ratios that mean the same thing.

*3:4 vs. 2:5 — which has more of the first ingredient per unit?***Compare**

To look at ratios and see which is bigger, smaller, or equal.

*8:12 → divide both by 4 → 2:3***Simplify**

To make a ratio as small as possible while keeping the same comparison.

*To compare  $\frac{3}{5}$  and  $\frac{4}{7}$ , use LCD 35:  $\frac{21}{35}$  vs  $\frac{20}{35}$* **Common denominator**

A bottom number that two fractions can share so you can compare them.

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can compare ratios by finding unit rates or using equivalent ratios.**

#### 1 Notice

Chef Academy is holding a Flavor Challenge!

#### 2 Key idea

I can compare ratios by finding unit rates or using equivalent ratios.

#### 3 Words I'll use

Unit rate

Equivalent ratio

Compare

Simplify

Common denominator



#### Think About It

- What quantities are being compared in each recipe?
- Can you tell just by looking which recipe has more cocoa per ounce?
- What would make the comparison fair?



#### Watch

Watch your teacher model one example. Jot what you see.

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#### We try

Solve the next one together as a class.

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#### You try

Now try one on your own.

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**Chef Reyes uses 3 tbsp cocoa for every 5 oz of milk, and Chef Tran uses 4 tbsp for every 7 oz. Can you tell whose cocoa is more chocolatey just by looking at the numbers? Why or why not?**

#### Level 1 support

**Start here:** You can't compare fairly until the ratios share the same amount of milk or you find cocoa per 1 oz.

#### Need a hint?

**Sentence starters (optional)**

**You \_\_\_\_ tell just by looking because the \_\_\_\_ amounts are different.**

*\_\_\_\_ se puede saber solo mirando porque las cantidades de \_\_\_\_ son diferentes.*

**To compare fairly we need the same \_\_\_\_ or the cocoa per \_\_\_\_.**

*Para comparar de forma justa necesitamos la misma \_\_\_\_ o el cacao por \_\_\_\_.*

**Word bank:** ratio compare unit rate equivalent ratio fair

**Listen for:** Students recognize that 3:5 and 4:7 have different milk amounts, so a fair comparison needs equal milk (a common amount) or a unit rate (cocoa per 1 oz).

#### Level 2

**Some students guess Chef Tran is more chocolatey because 4 is more than 3. Why is that reasoning not enough? Justify using the milk amounts.**

- Looking only at the cocoa is not enough because \_\_\_\_.
- More cocoa does not always mean more chocolatey since \_\_\_\_.

**EXPLORE**

**You scaled both recipes to 35 oz of milk: Chef Reyes became 21:35 and Chef Tran became 20:35. How does scaling to a common denominator let you compare the recipes fairly?**

**Level 1 support**

**Start here:** When both ratios share 35 oz of milk, you only have to compare the cocoa amounts.

 **Need a hint?**

**Sentence starters (optional)**

**After scaling to 35 oz, Chef Reyes has \_\_\_\_ tbsp and Chef Tran has \_\_\_\_ tbsp.**

*Después de escalar a 35 oz, Chef Reyes tiene \_\_\_\_ cucharadas y Chef Tran tiene \_\_\_\_ cucharadas.*

**Now I can compare because the \_\_\_\_ is the same, so I just compare the \_\_\_\_.**

*Ahora puedo comparar porque la \_\_\_\_ es igual, así que solo comparo el \_\_\_\_.*

**Word bank:** **equivalent ratio** **common denominator** **compare** **simplify** **milk**

**Listen for:** Students explain that equal milk (35 oz) makes the comparison fair, so  $21 \text{ tbsp} > 20 \text{ tbsp}$  means Chef Reyes is more chocolatey.

**Level 2**

**Find the unit rate (cocoa per 1 oz) for each chef and show it gives the same winner as the 35 oz scaling. Why must both methods agree?**

- The unit rates are about \_\_\_\_ and \_\_\_\_ tbsp per oz, so \_\_\_\_ wins.
- Both methods agree because \_\_\_\_.

CONNECT

**Two pizza shops have deals: Mario's sells 2 large pizzas for \$18, and Sal's sells 3 for \$25. How would you compare these to find the best price per pizza?**

Level 1 support

**Start here:** Find the unit rate (price per 1 pizza) for each shop, then compare those rates.



**Need a hint?**

**Sentence starters (optional)**

**Mario's costs \_\_\_\_ per pizza because 18 divided by 2 equals \_\_\_\_.**

*Mario's cuesta \_\_\_\_ por pizza porque 18 entre 2 es \_\_\_\_.*

**The better deal is \_\_\_\_ because its \_\_\_\_ is lower.**

*La mejor oferta es \_\_\_\_ porque su \_\_\_\_ es más baja.*

**Word bank:** unit rate compare ratio better deal per

**Listen for:** Students compute \$9.00/pizza for Mario's and about \$8.33/pizza for Sal's, then choose Sal's because the lower unit rate is the better deal.

Level 2

**If you only need 2 pizzas, does the better unit rate always mean the better total price?**

**Explain when buying the deal with the lower unit rate might not save money.**

- A lower unit rate is the best total deal only when \_\_\_\_.
- It might not save money if \_\_\_\_.

## Guided Examples

### Example 1

**Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?**

**Solution:** Chef A:  $2 \div 8 = 0.25$  cups per cracker. Chef B:  $3 \div 9 \approx 0.33$  cups per cracker. Chef B uses more cheese per cracker.

**Answer:** A. Chef B

### Example 2

**Which ratio is greater: 3:4 or 5:8?**

**Solution:** Convert to the same denominator:  $3:4 = 6:8$ . Since  $6:8 > 5:8$ , the ratio 3:4 is greater.

**Answer:** A. 3:4

### Example 3

**Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.**

**Solution:** Recipe A:  $2 \div 6 \approx 0.33$  lemons per cup. Recipe B:  $3 \div 7 \approx 0.43$  lemons per cup. Since  $0.43 > 0.33$ , Recipe B is more lemony.

**Answer:** A. Recipe B

## Write About the Math

### The Writing Revolution

I can explain my comparison using the words unit rate, equivalent ratio, compare, and common denominator.

#### 1. Kernel Sentence subject + verb

**Model:** Compare is to look at ratios and see which is bigger, smaller, or equal.

*Comparar es mirar razones para ver cuál es mayor, menor o igual.*

**Write a kernel sentence about compare. Use a subject and a verb.**

*Escribe una oración base sobre comparar. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because · but · so

**Kernel:** Compare matters in math

*Comparar importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Compare matters in math because \_\_\_\_.**

*Comparar importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Compare matters in math, but \_\_\_\_.**

*Comparar importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Compare matters in math, so \_\_\_\_.**

*Comparar importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

#### Statement *Afirmación*

Tell one true fact about compare.  
*Di un hecho verdadero sobre compare.*

**Compare \_\_\_\_.**

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#### Question *Pregunta*

Ask a question about compare.  
*Haz una pregunta sobre compare.*

**How does \_\_\_\_ ?**

*¿Cómo \_\_\_\_ ?*

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#### Exclamation *Exclamación*

Show excitement about compare.  
*Muestra entusiasmo sobre compare.*

**Wow, \_\_\_\_ !**

*¡Guau, \_\_\_\_ !*

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#### Command *Mandato*

Tell a partner what to do with compare.  
*Dile a un compañero qué hacer con compare.*

**First, \_\_\_\_ .**

*Primero, \_\_\_\_ .*

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### 4. Explain Your Reasoning use a sentence starter

**You \_\_\_\_ tell just by looking because the \_\_\_\_ amounts are different.**

*\_\_\_\_ se puede saber solo mirando porque las cantidades de \_\_\_\_ son diferentes.*

**To compare fairly we need the same \_\_\_\_ or the cocoa per \_\_\_\_.**

*Para comparar de forma justa necesitamos la misma \_\_\_\_ o el cacao por \_\_\_\_.*

**After scaling to 35 oz, Chef Reyes has \_\_\_\_ tbsp and Chef Tran has \_\_\_\_ tbsp.**

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Store A sells 3 mangoes for \$6. Store B sells 5 mangoes for \$9. Which store offers a better price per mango?**

- A. Store B
- B. Store A
- C. Same price
- D. Cannot determine

Show your work:

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**2. Store A sells 5 notebooks for \$12. Store B sells 8 notebooks for \$20. Without calculating, a student says Store B must be cheaper because you get more notebooks. Is the student's reasoning valid? Find the unit rates to prove your answer.**

Show your work:

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### Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**Find Kai's Mistake — find the error, then write the correct reasoning.**

Show your work:

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## Reflect — Exit Ticket

**A bakery sells 4 muffins for \$10 and another bakery sells 6 muffins for \$14. Which bakery has the lower price per muffin?**

- A. The second bakery
- B. The first bakery
- C. Same price
- D. Cannot determine

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** A. Store B — *Store A:  $\$6 \div 3 = \$2.00$  per mango. Store B:  $\$9 \div 5 = \$1.80$  per mango. Store B has the better (lower) price per mango.*
2. **Try It 2:** The student's reasoning is not valid — getting more items does not mean a lower price per item. Store A:  $\$12 \div 5 = \$2.40$  per notebook. Store B:  $\$20 \div 8 = \$2.50$  per notebook. Store A is actually cheaper per notebook, even though Store B offers more notebooks per purchase.
3. **Exit Ticket:** A. The second bakery — *First bakery:  $\$10 \div 4 = \$2.50$  per muffin. Second bakery:  $\$14 \div 6 \approx \$2.33$  per muffin. The second bakery has the lower price.*

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Compare is to look at ratios and see which is bigger, smaller, or equal.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).